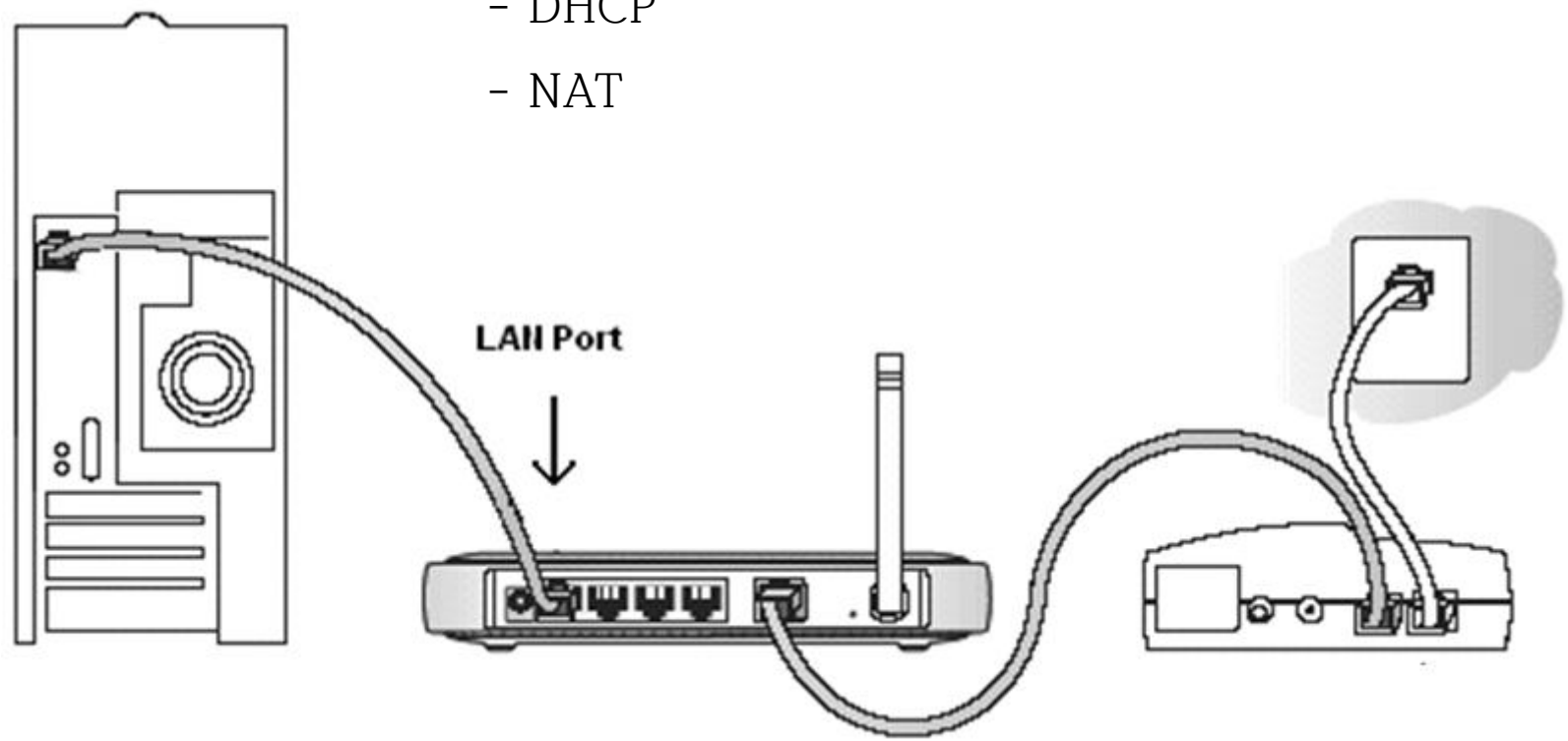
# NAT

# SoHo(Small office/home office)에서의 NAT

- 공유기 기능
  - Routing
  - DHCP
  - NAT




❶ 컴퓨터 ❷ 유/무선 공유기 ❸ 모뎀

# NAT 기능

- 네트워크 주소 변환 (Network Address Translation)
- 송신지 또는 수신지 IP address를 다른 주소로 변환
- Source NAT(SNAT)와 Destination NAT(DNAT)

| 송신지 IP | 수신지 IP |
|---|---|
| **192.168.1.10** | 200.10.10.10 |

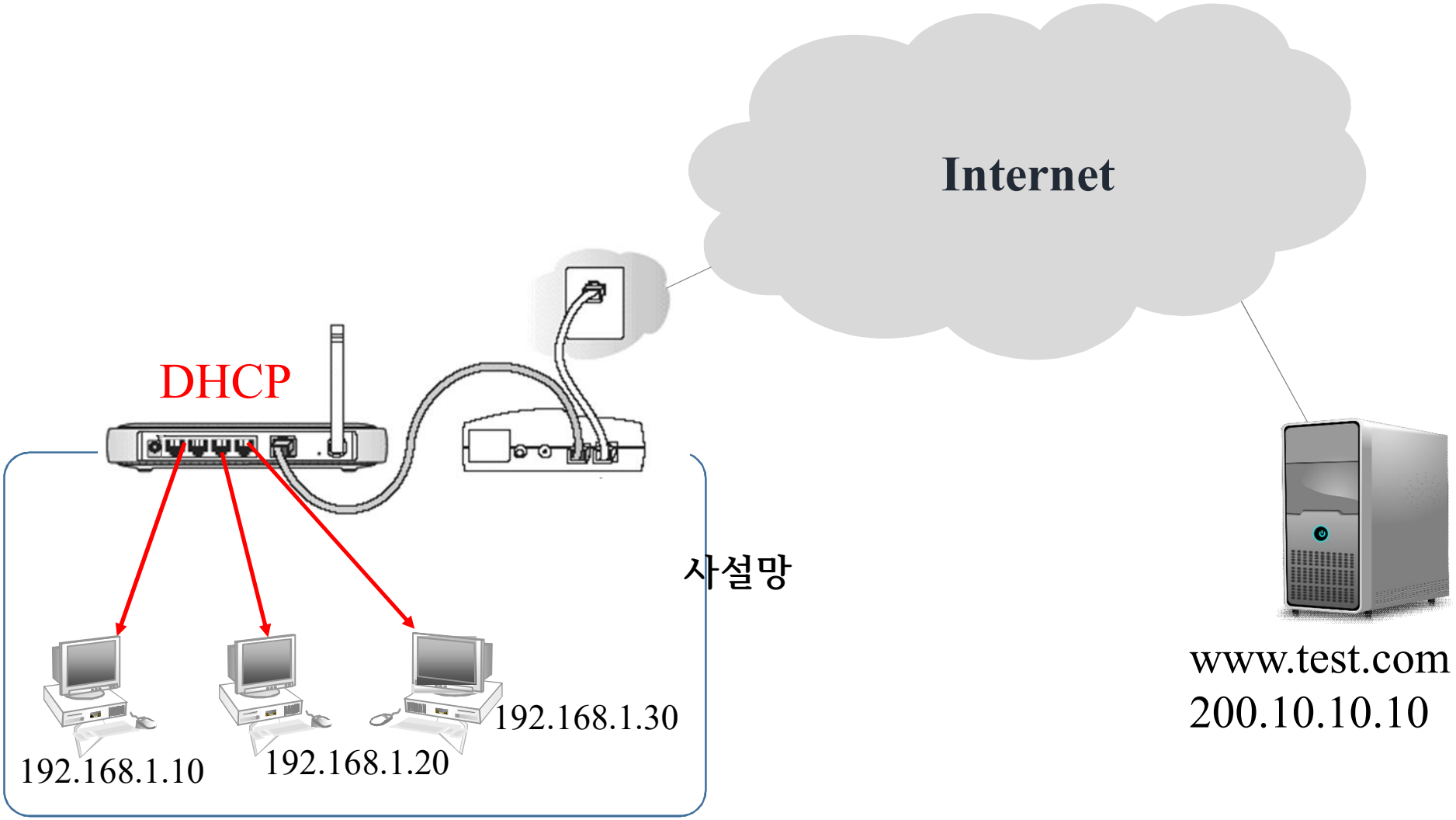
❶ **SNAT**

| | |
|---|---|
| **172.16.1.10** | 200.10.10.10 |

| 송신지 IP | 수신지 IP |
|---|---|
| 192.168.1.10 | **200.10.10.10** |

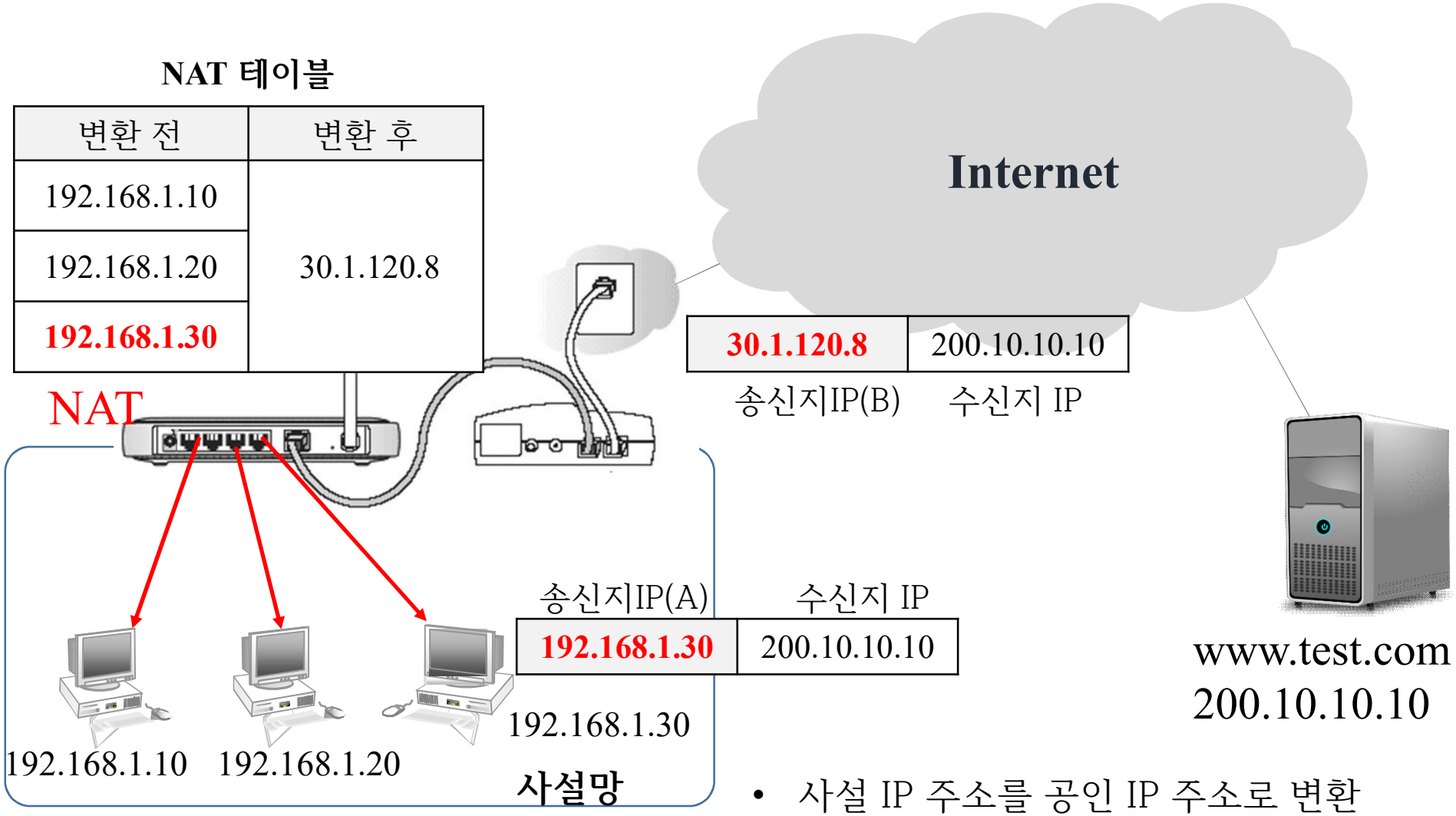
❷ **DNAT**

| | |
|---|---|
| 192.168.1.10 | **10.10.10.10** |


Internet
DHCP
사설망
192.168.1.10
192.168.1.20
192.168.1.30
www.test.com
200.10.10.10

**NAT 테이블**

| 변환 전 | 변환 후 |
|---|---|
| 192.168.1.10 | 30.1.120.8 |
| 192.168.1.20 | |
| **192.168.1.30** | |



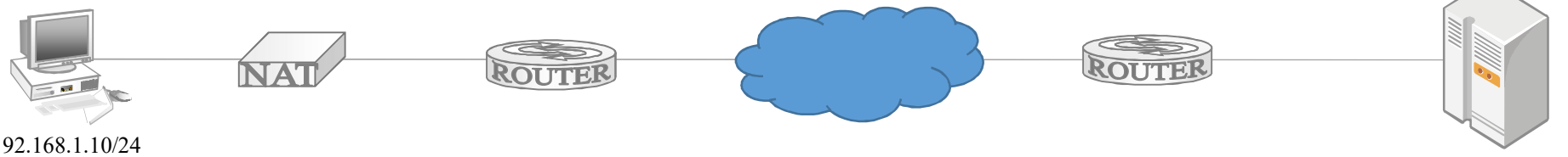

- 사설 IP 주소를 공인 IP 주소로 변환

# Source NAT

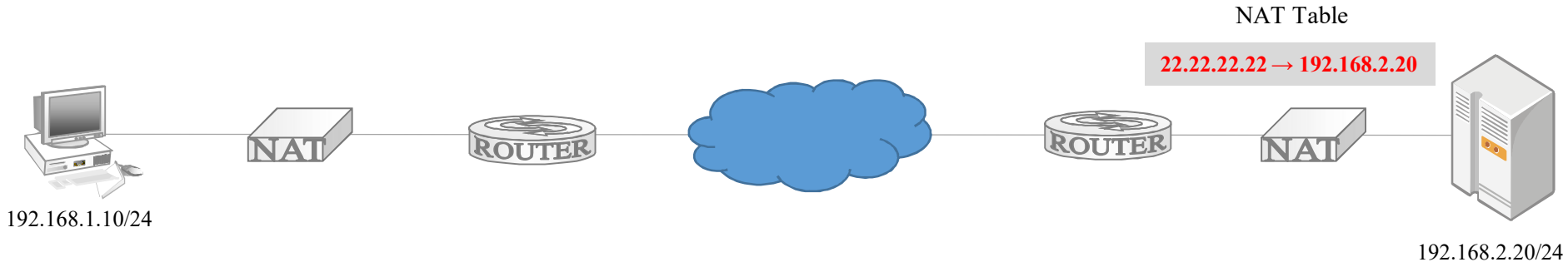
NAT Table

192.168.1.10 → 11.11.11.11

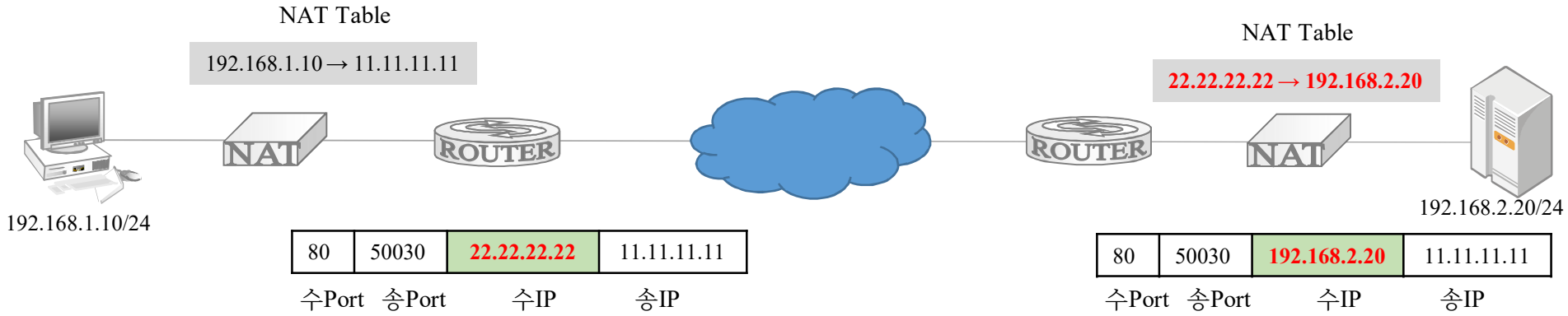
| 수Port | 송Port | 수IP | 송IP |
|---|---|---|---|
| 80 | 50030 | 22.22.22.22 | 11.11.11.11 |



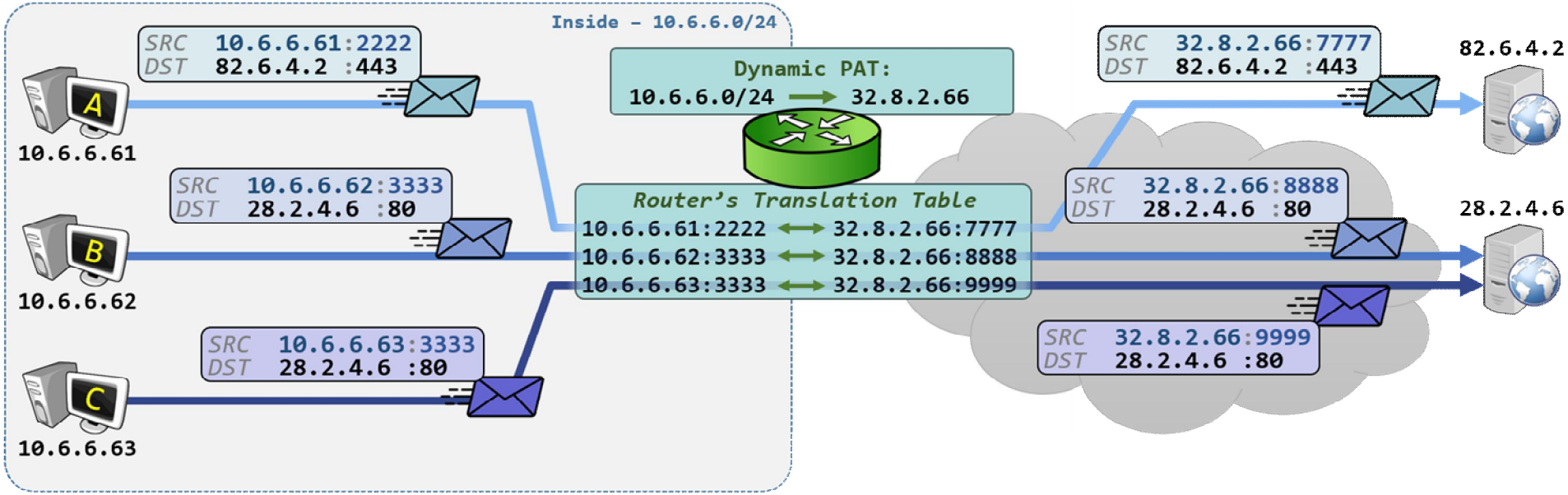

| 수Port | 송Port | 수IP | 송IP |
|---|---|---|---|
| 80 | 50030 | 22.22.22.22 | 192.168.1.10 |

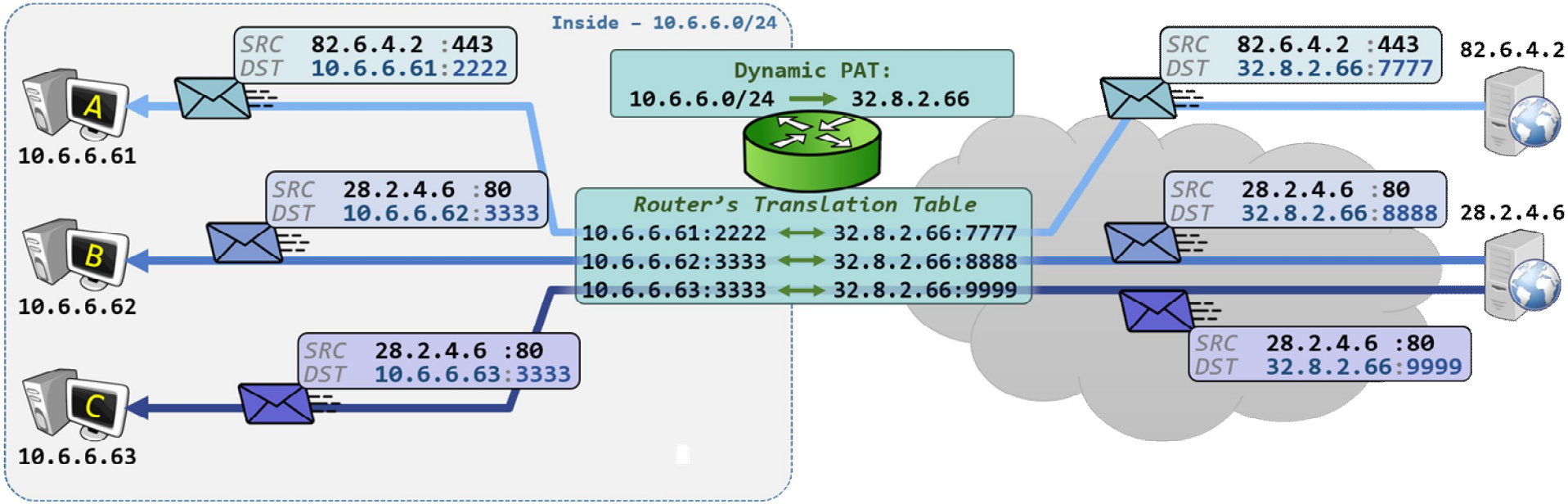



# Destination NAT



# PAT


Inside – 10.6.6.0/24
SRC 10.6.6.61:2222
DST 82.6.4.2 :443
A
10.6.6.61
SRC 10.6.6.62:3333
DST 28.2.4.6 :80
B
10.6.6.62
SRC 10.6.6.63:3333
DST 28.2.4.6 :80
C
10.6.6.63
Dynamic PAT:
10.6.6.0/24 → 32.8.2.66
Router's Translation Table
10.6.6.61:2222 ↔ 32.8.2.66:7777
10.6.6.62:3333 ↔ 32.8.2.66:8888
10.6.6.63:3333 ↔ 32.8.2.66:9999
SRC 32.8.2.66:7777
DST 82.6.4.2 :443
SRC 32.8.2.66:8888
DST 28.2.4.6 :80
SRC 32.8.2.66:9999
DST 28.2.4.6 :80
82.6.4.2
28.2.4.6


Inside – 10.6.6.0/24
SRC 82.6.4.2 :443
DST 10.6.6.61:2222
Dynamic PAT:
10.6.6.0/24 → 32.8.2.66
SRC 82.6.4.2 :443
DST 32.8.2.66:7777
82.6.4.2
A
10.6.6.61
SRC 28.2.4.6 :80
DST 10.6.6.62:3333
Router's Translation Table
10.6.6.61:2222 ↔ 32.8.2.66:7777
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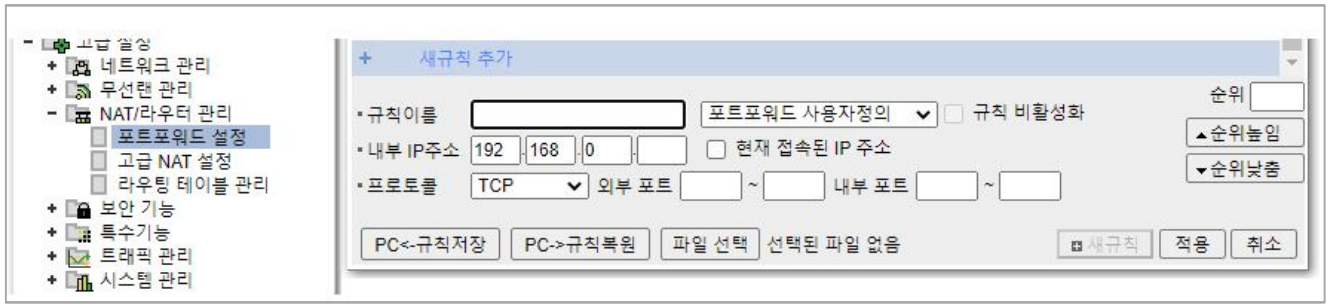
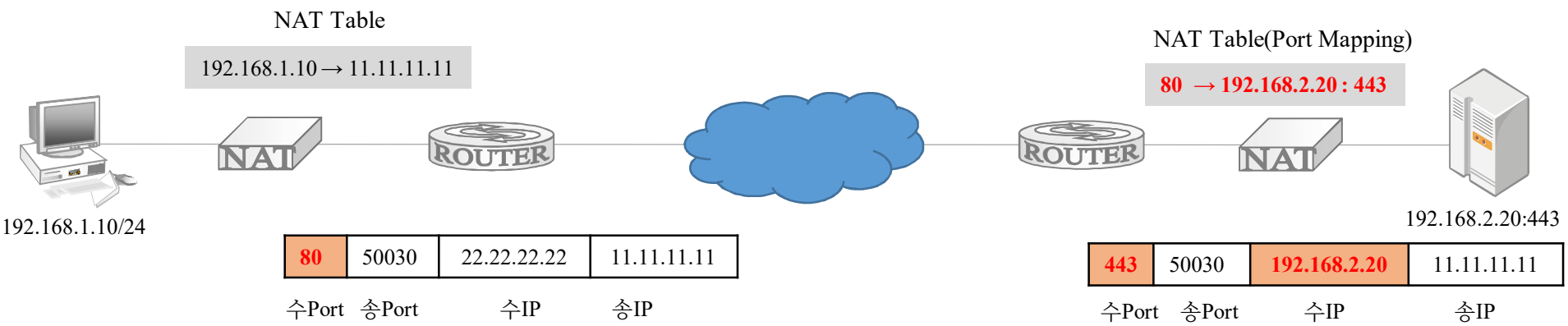
DST 32.8.2.66:8888
28.2.4.6
B
10.6.6.62
SRC 28.2.4.6 :80
DST 10.6.6.63:3333
SRC 28.2.4.6 :80
DST 32.8.2.66:9999
C
10.6.6.63

# NAT Port Mapping (Port Forwarding)



http://200:1.1.10

**HTTP**

**192.168.1.10:80**

**FTP**

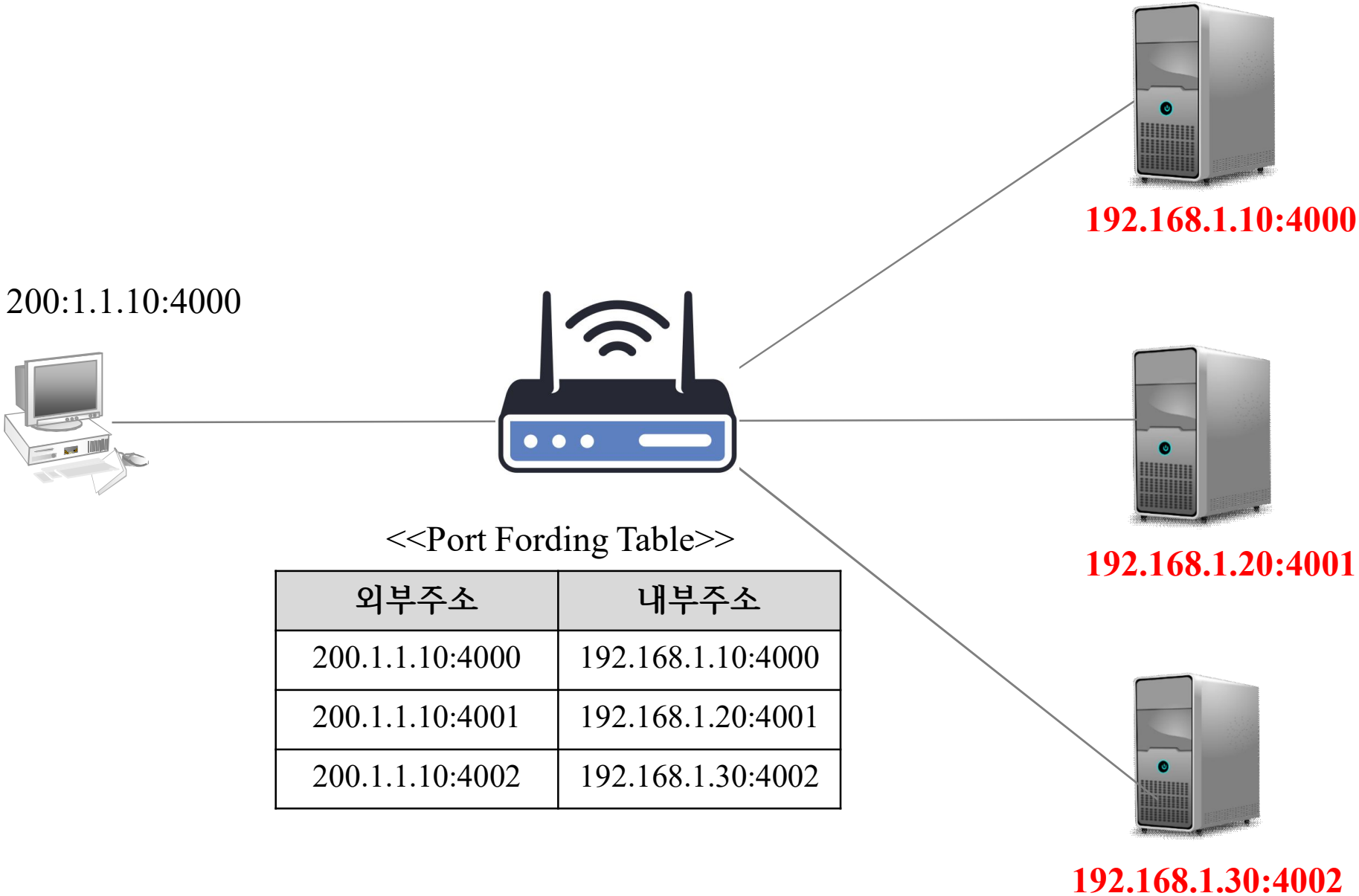
**192.168.1.20:21**

**RDP**

**192.168.1.30:3389**

<<Port Fording Table>>

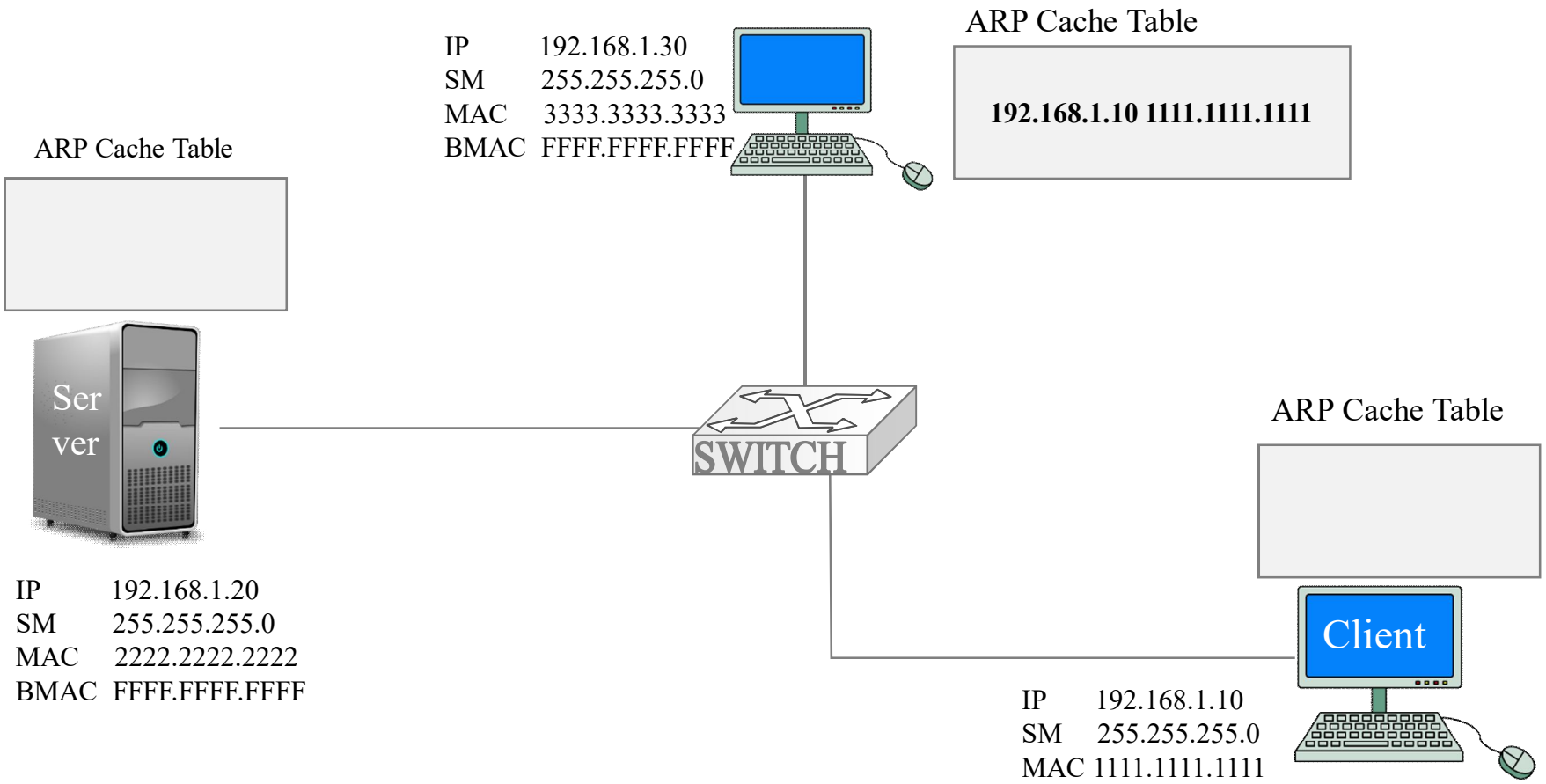
| 외부주소 | 내부주소 |
|---|---|
| 200.1.1.10:80 | 192.168.1.10:80 |
| 200.1.1.10:21 | 192.168.1.20:21 |
| 200.1.1.10:3389 | 192.168.1.30:3389 |




<<Port Fording Table>>

| 외부주소 | 내부주소 |
|---|---|
| 200.1.1.10:4000 | 192.168.1.10:4000 |
| 200.1.1.10:4001 | 192.168.1.20:4001 |
| 200.1.1.10:4002 | 192.168.1.30:4002 |

# ARP cache table

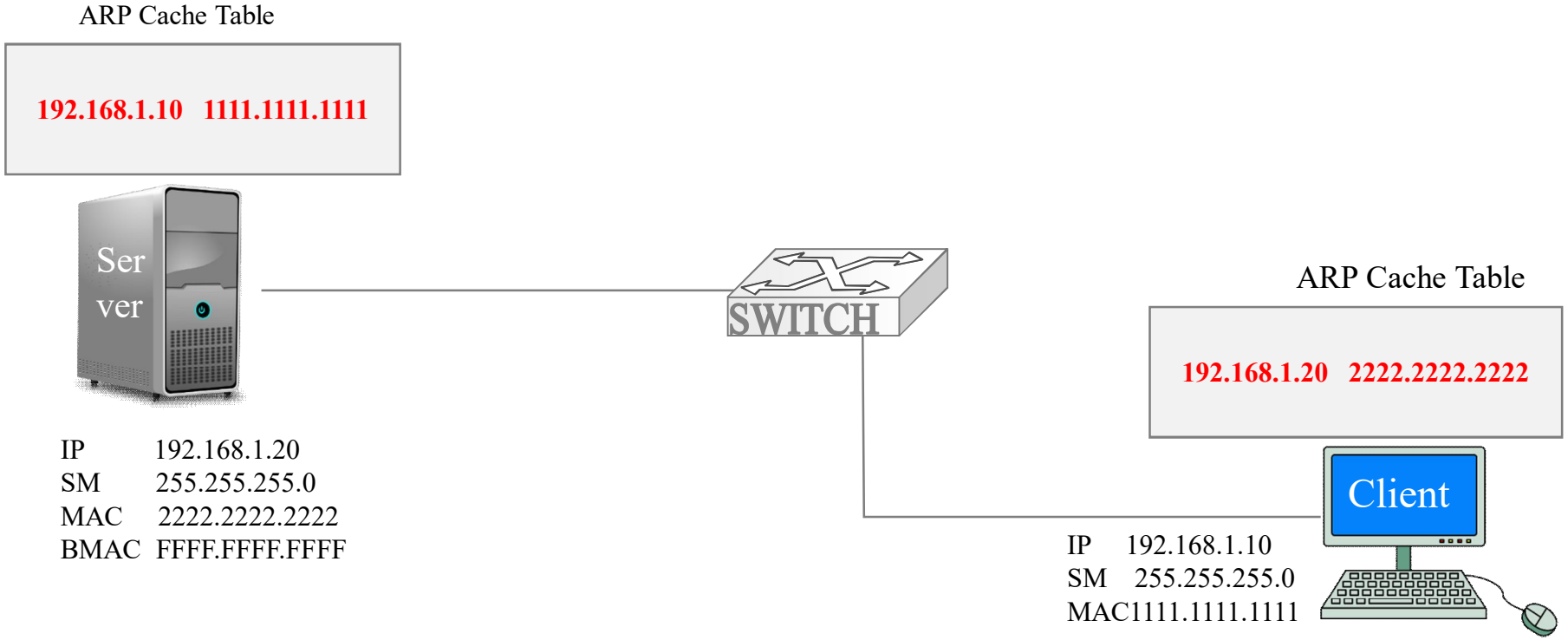



ARP Request

| 수MAC | 송MAC | Protocol | ARP Header | |
|---|---|---|---|---|
| FFFF.FFFF.FFFF | 1111.1111.1111 | 0x0806 | 1111.1111.1111 /192.168.1.10<br>**0000.0000.0000/192.168.1.20** | 송신지<br>수신지 |

| 수MAC | 송MAC | 송IP | 수IP | 송Port | 수Port | 전송데이타 |
|---|---|---|---|---|---|---|
| **????** | 1111. 1111.1111 | 192.168.1.10 | **192.168.1.20** | 50030 | 80 | Get /www.test.com |

| | ARP Header | Protocol | 송MAC | 수MAC |
|---|---|---|---|---|
| 송신지<br>수신지 | 2222.2222.2222 /192.168.1.20<br>1111.1111.1111 /192.168.1.10 | 0x0806 | 2222.2222.2222 | 1111.1111.1111 |




| 수MAC | 송MAC | 송IP | 수IP | 송Port | 수Port | 전송데이타 |
|---|---|---|---|---|---|---|
| 222.2222.2222 | 1111. 1111.1111 | 192.168.1.10 | 192.168.1.20 | 50030 | 80 | Get /www.test.com |